# 1 Once-counted reduced four-block cancellation

Let $U_N$ be unitary, $P_{xy} = |U_{xy}|^2$, and $J_{xy} = N^{-1}$. Assume, eventually, $\max P_{xy} \le C/N$ and $\|P^L - J\| \le 2^{-L}$ for every integer $L \ge 0$. BGS-FM-021 supplies these bounds for the unchanged frozen map. Let $T_m^{(4)}$ be the set of rooted ordered coordinate-word pairs obtained by cutting the first cyclic word into four nonempty blocks $A, B, C, D$, whose initial vertex values are pairwise distinct, and taking a cyclic rotation of $ADCB$ for the second word. Each pair is counted once, regardless of its witnesses. This is the unique reduced four-block ordering: all four original successor boundaries change. Put $\mathcal{A}(x) = \prod_{j \bmod m} U_{x_j x_{j+1}}$.

**Theorem (BGS-TRACE-016).** For every smooth $w$ compactly supported in $(0, \infty)$,

$$\frac{1}{N^2} \sum_m w(m/N) \sum_{(x,y) \in T_m^{(4)}} \mathcal{A}(x)\overline{\mathcal{A}(y)} \longrightarrow 0.$$

For $K = \lceil \log N \rceil$ and support in $[a, b] \subset (0, \infty)$, the absolute value of this complete signed band sum is bounded by

$$O_w(K^{10}/N) + O_w\Big(N^{10}(C/N)^{K/2}\Big) + O_w\Big(N^5 2^{-(aN-4)/4}\Big).$$

The startup inherited from the mixing theorem is not effective.

**Marked cancellation and exact correction.** The marked four-link sum $M_m$ has two independent trace-root factors. Replacing any one link by $J$ gives zero by column orthogonality and distinct ports, so its longest link makes $M_m$ exponentially small. The local overlap $O_m$ restricts the first two blocks to $A = (a)$ and $B = (b, a, b)$, with arbitrary nonempty $C, D$. It retains a free long endpoint: uniform $C$ gives $\delta_{db}/N = 0$, and uniform $D$ gives $\delta_{ac}/N = 0$. Its normalized sum is likewise exponentially small. The once-counted correction is $M_m/4 - O_m$, up to the stated exceptional error; a uniform multiplicity division is not imposed on all pairs.

**Combinatorial control.** A word with distinct length-$K$ cyclic substrings localizes every competing cut within $O(K)$ of an original cut. Indeed a common refinement interval of length at least $(r+1)K$ fixes its translation offset. Each mismatch component begins at a common genuine cut, since otherwise two cut heads would have equal vertex values. A longest common interval fixes cyclic alignment, without an extra factor $m$. Three fixed anchors in different original four-block pieces force cut uniqueness by prefix cancellation. The corrected argument handles all three locations of the unanchored block, including empty prefixes and equal lengths. Thus ambiguity requires two short original blocks. There are $O(K^2 m)$ original length choices and $O(K^4)$ competing cut sets. Exposing all four starts first pays every equality-rank constraint by a transition bounded by $C/N$, giving $O(K^6/N)$ for rank at least three.

The separately proved computer-assisted classification identifies every rank-two relation with the local flip $(s\ s+2)(s+1\ s+3)$ on an alternating four-vertex cluster. Its finite kernel is certified separately by `Bgs/FourCutKernel.lean`; the universal compression and the analytic argument here are not formalized by that bounded certificate. Two distinct local flips compose to rank four, except neighboring flips, whose product is the forbidden three-cycle case. The representation graph is therefore triangle-free. A pair without a higher-rank relation has one or two cut sets; subtracting its single local edge counts it once. Higher multiplicities cost at most another $O(K^4)$, yielding $O(K^{10}/N)$. Repeated length-$K$ substrings, including periodic words, have superpolynomially small exceptional marked absolute mass.

**Boundary.** This is a fixed-family theorem under generic unitarity, flatness and position-chain mixing hypotheses. It is compatible with nonuniversal spectra satisfying those bounds. Repeated ports, non-permutation pairs, overlaps among families, higher or growing reconnection complexity, plateau restoration and the full signed hierarchy remain open. The rounding obstruction precludes inferring the exact spectral target from these coarse controls alone. No deterministic BGS conclusion follows.